

Seeing is Feeling: A Novel Haptic Display for Wearer-Observer Mutual Haptic Understanding

Arata Horie

Research Center for Advanced Science and Technology
The University of Tokyo
Tokyo, Japan
arata.horie@star.rcast.u-tokyo.ac.jp

Ryo Murata

Graduate School of Information Science and Technology
The University of Tokyo, Tokyo, Japan
Tokyo, Japan
murata@star.rcast.u-tokyo.ac.jp

Zendai Kashino

Research Center for Advanced Science and Technology
The University of Tokyo, Tokyo, Japan
Tokyo, Japan
kashino@star.rcast.u-tokyo.ac.jp

Masahiko Inami

Research Center for Advanced Science and Technology
The University of Tokyo, Tokyo, Japan
Tokyo, Japan
inami@star.rcast.u-tokyo.ac.jp



Figure 1: (a)The device is a wearable haptic presentation-sharing display (b)The device consists with haptic presentation and exaggeration part (c)The system provides interaction with mutual haptic understanding between the wearer and observer.

ABSTRACT

We propose a new haptic display that enables mutual understanding of haptic sensation between the wearer and an observer. In addition to presenting haptic sensations by inducing skin deformation, we have achieved creating a mutual understanding of the sensations with the observer by making the haptic stimulus evident. People have the ability to predict the sensations of others by observing their sensory states. The system is composed of a part that provides haptic stimulus while creating visible skin deformation, and a mechanical structure that visually exaggerates the deformation. The proposed system realizes richer interactions by extending the entertainment experiences of the observers during live content, increase understanding of internal states through biofeedback, and be used in remote haptic communication.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

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KEYWORDS

haptics, skin deformation, haptic sharing, wearable device

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1 INTRODUCTION

A mutual understanding of haptic sensations between individuals is typically achieved through a mechanical exchange of touch. By touching another individual, one can feel the touch that the other individual is feeling by feeling how the other's skin pushes back. The ability to interpret this reaction force to know what the other person is feeling is a function that we acquire through daily experience of touch. However, contact is not always necessary to achieve a mutual understanding of the sense of touch.

Several studies have shown that humans have the ability to estimate the sensory state of others by visually observing them [Keyzers et al. 2004; Preston and De Waal 2002]. Keyzers et al. confirmed that the somatosensory cortex of an observer is activated by watching a video of another person's foot being stroked [Keyzers et al. 2004]. Therefore, visualization of the haptic state of others may enable a mutual understanding of the sense of touch even without contact.

Conventional haptic display were build without considering how they would be seen, so observers could not estimate the user's sensory state[Horie et al. 2020]. Several haptic display which are designed to share the haptic experience with an observer, have typically used LEDs[Konishi et al. 2016]. While these are able to represent the experience, an observer cannot be sure what kind of haptic sensation is being presented, or even whether the LEDs signify a haptic sensation, without explanation.

We propose a wearable system that presents haptic sensation to the wearer while making the haptic sensation evident to outside observers as shown in Fig.1(a). The sensations are made evident by making the skin deformations clearly visible to observers. By using the proposed system, a wearer can feel haptic sensations that an observer can intuitively understand. We also describe prototypes with an additional mechanism that exaggerates the sensory intensity estimated by the observer while still guaranteeing that the haptic sensation presented to the wearer is evident. Finally, we introduce user experiences that becomes possible by using this prototype.

2 SYSTEM

The proposed system shares the sense of touch with observers by making the presented sense of touch evident. This system leverages the human ability to estimate sensory states through observation.

The system presents haptic sensations of arbitrary intensity to the wearer and guarantees, to an observer, that they are being presented. First, the tactor pressed into the skin to present the haptic sensation. Then, the observer can directly observe how much the tactor is being pushed into the skin. The observer can then estimate the intensity of the haptic sensation perceived by the wearer. We have confirmed that people are able to make accurate predictions in this manner through psychophysical experiments.

The system also has a mechanical structure to amplify the motion and exaggerate the skin deformation. The deformation of the human skin is only a few millimeters at most, and it becomes difficult to estimate the deformation after a small distance. Our system has a mechanical structure that interacts with the tactor, such that the deformation of the skin can be transmitted to others in an exaggerated form as shown in Fig.1(b). By using a simple mechanical structure without additional actuators, the system can guarantee that motion of the structure is directly related to skin deformation.

The system is capable of presenting not only strength (pressure), but also spatial distributions of strength by connecting multiple modules. This means that we can design stimuli that make the wearer perceive spatial tactile flow and transmit it to the observer. Namely, the system can be used to present a smooth Moving Phantom Sensation. We have identified the parameters that allow us to perceive directional flow visually and in a haptic manner, and have incorporated them into the device control algorithm.

3 USER EXPERIENCE

3.1 Sharing an Avatar's Haptic Sensations

Presenting haptic sensations presented to an avatar in a virtual space is a common application of haptic displays. The proposed system can not only present the wearer with the haptic sensations, but also share the experience with another observer as shown in

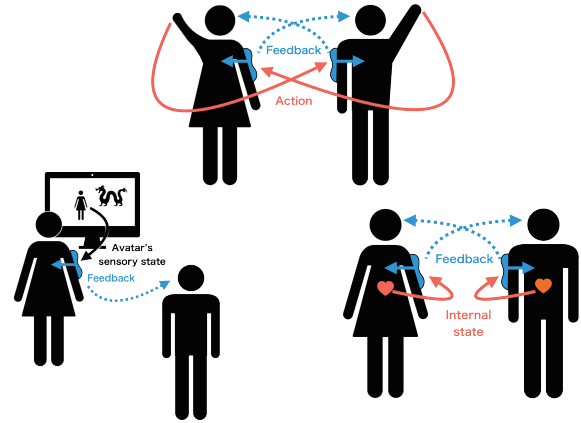


Figure 2: The proposed system enable a variety of haptic interactions involving wearers and observers.

Fig.2(Left). By using the proposed system, the observer can intuitively understand the haptic sensation presented to the wearer. It can be applied to entertainment involving both a primary player and an external observer, such as live streamed content.

3.2 Sharing the Internal State of the Body

Biofeedback is used to understand the internal state of one's own body. By using the proposed system, a user can become aware of the internal state of their own body and simultaneously communicate it to others. By driving the system to synchronize with respiration, heartbeat, muscle activity, etc., it can be applied to entertainment with internal states and body motion instruction as shown in Fig.1(c) and Fig.2(Right) .

3.3 Remote Haptic Communication

Communication through the sense of touch cannot be established without knowing the sensation one is giving to the other. This system allows us to be sure that our actions are affecting others the way we intend as shown in Fig.2(Top). It allows individuals to reach a mutual understanding of what is being sensed, as in normal haptic communication.

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